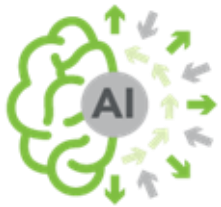
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AI CLUSTERS




# Customer-Contextual Fault Management for Bare Metal GPU Infrastructure

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# Agenda

- GPU Infrastructure Availability Challenges
- The Data from Behind the Scenes
- Customer-Collaborative Fault Management Architecture Overview
- Lessons Learned & Call to Action





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# The Challenges

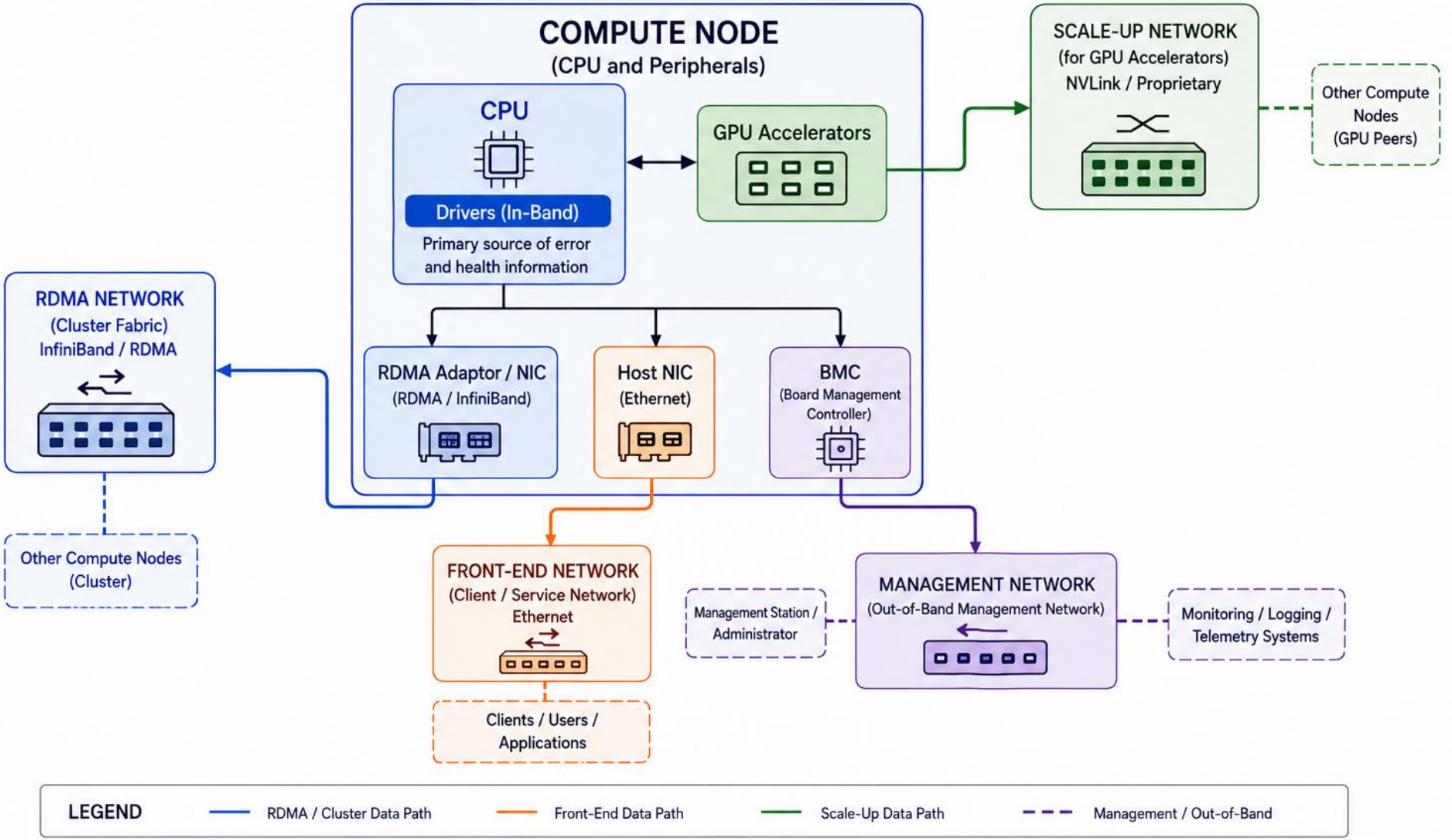
- GPU bare-metal platforms are becoming increasingly complex, but observability has not kept pace.
  - Telemetry asymmetry
    - Visibility gap between what the customer observes and what hyper-scalar can know.
  - Slow path to recovery
    - Repeat repairs, longer recovery times, and reduced platform availability.
  - Security constraints
    - Security policies often prevent in-band agents, creating an observability gap where key failure signatures are visible to customers but not to hyperscalers.
- New offerings introduce a unique split of observability and availability responsibilities between the cloud provider and the customer.
    - **Pros:** Provides greater isolation of the customer's environment for maximum control and performance
    - **Cons:** Limiting cloud provider access to key telemetry for triage and repair.


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# GPU Bare-Metal Compute Node and Peripheral Connectivity

*Peripherals connect to different network fabrics based on their function.*
*Error and health information is primarily exposed in-band via drivers running on the CPU.*

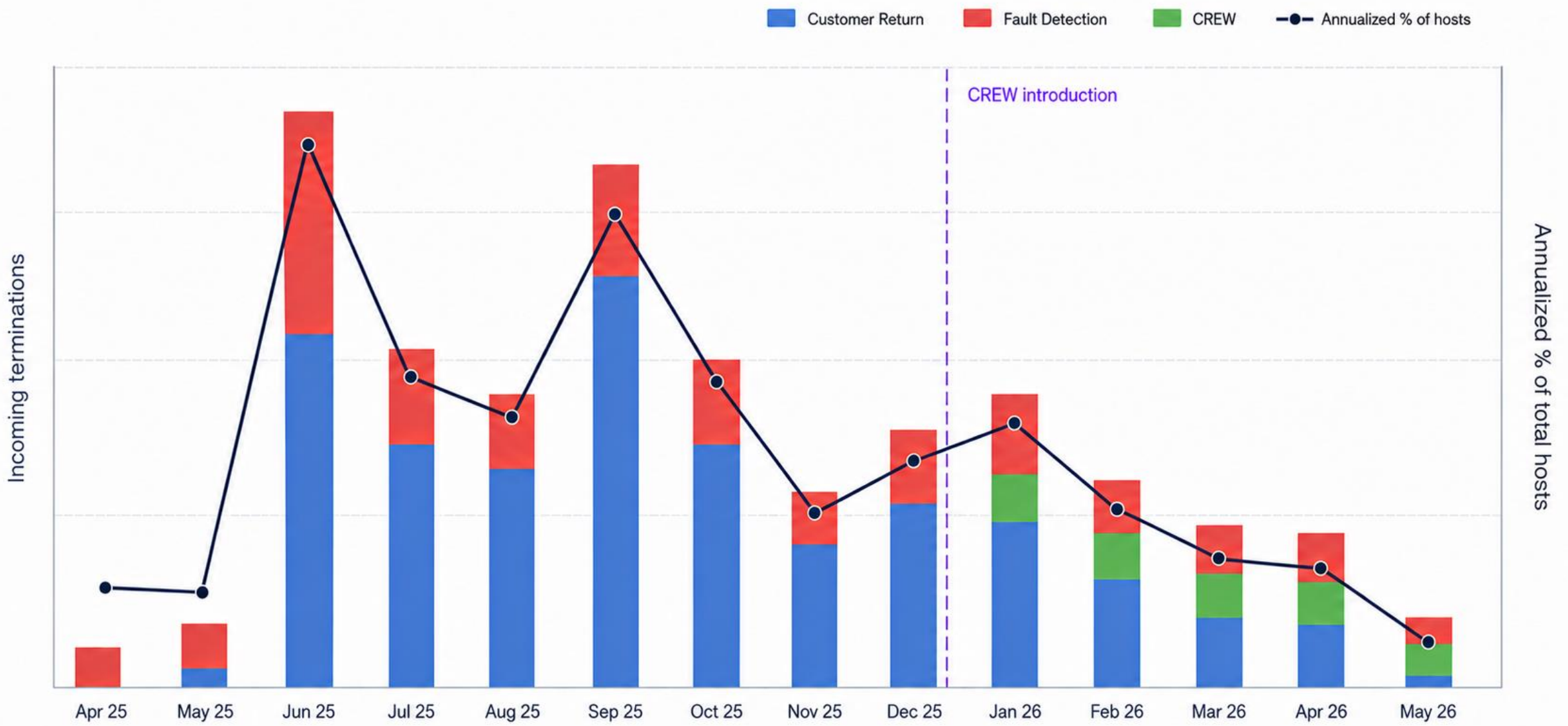



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# Incoming Terminations – GB200 Platform in One AD Deployment

Monthly incoming volume by source; annualized host rate on secondary axis




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# Overview: Customer collaborative failure management

- Various stages of CREW involve
  - **Report** : Customer submits a dedicated-host fault and diagnostic evidence through the ComputeReportedFaults API.
  - **Sanitize** : Evidence is inspected in an isolated OCI-internal sandbox; untrusted files never enter the substrate environment.
  - **Triage** : OCI service evaluates the sanitized evidence, classifies the fault, and determines whether hardware repair is required.
  - **Repair & Track** : CREW creates and links a MaintenanceEvent. Customers receive notifications and track the repair through existing Fault, Work Request, and Instance Maintenance APIs.
  - **Learn & Automate** : Learnings from CREW cases create a feedback loop that: -
    - Informs vendor discussions with field evidence and recurring failure patterns
    - Enhances the existing telemetry infrastructure to detect these conditions automatically
    - Progressively replaces customer-sourced diagnostics with proactive, platform-generated signals



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# Lessons learned and Call to Action

- Customer collaboration can close the observability gap.
  - Customer-provided in-band failure evidence, combined with OCI out-of-band telemetry, has shown strong improvement in failure diagnosis, remediation, and platform availability for GPU platforms.
- Standardization is the key to scaling the learning.
  - Leverage the OCP community to turn proven practices into common standards.
- Inviting collaborators to co-author an OCP white paper that captures learnings and explores a standardized, industry-wide approach to customer-driven failure diagnosis and remediation.
- Oracle is active participant in OCP workstreams, including:
  - OCP Hardware Management — RAS APIs and hardware fault management
  - GPU Management — telemetry, health, and failure management



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# Thank You!



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October 12-15, 2026
San Jose, California

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